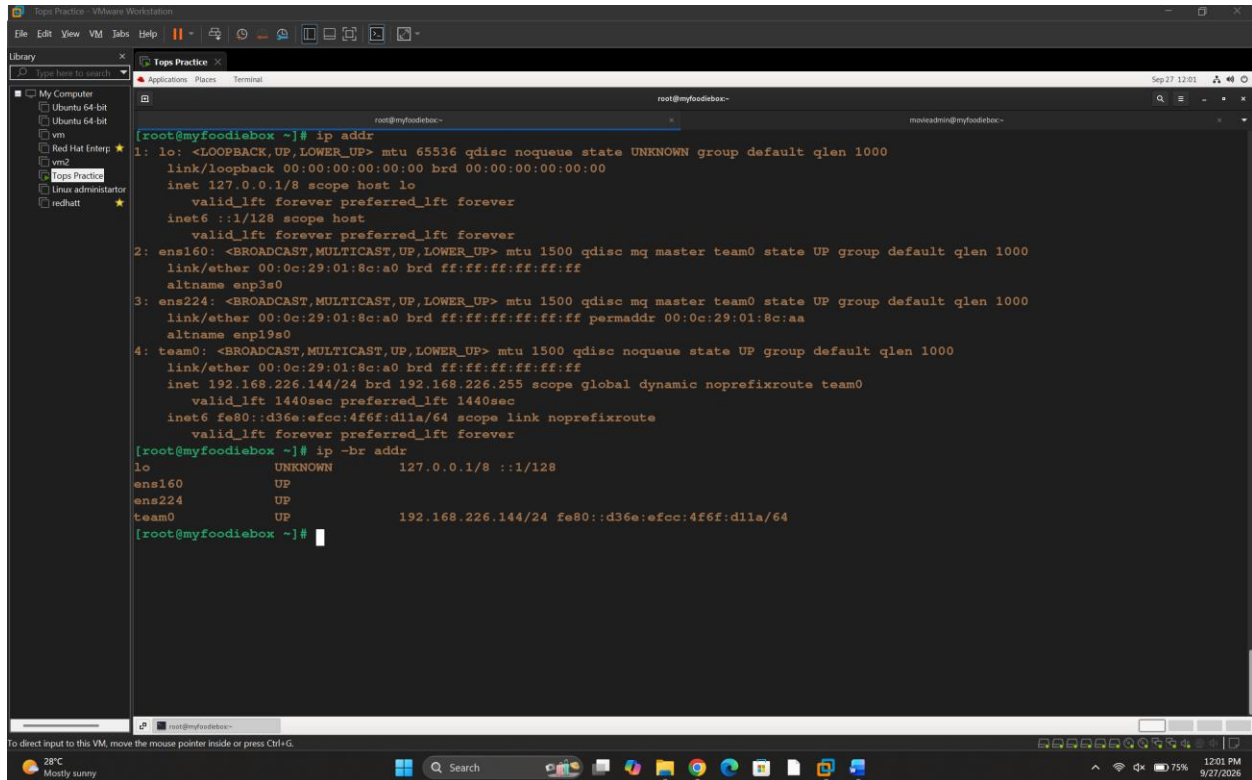


1.

Use the ifconfig or ip command to display the current IP address and network interfaces on your Linux system, then take a screenshot of the output.

>



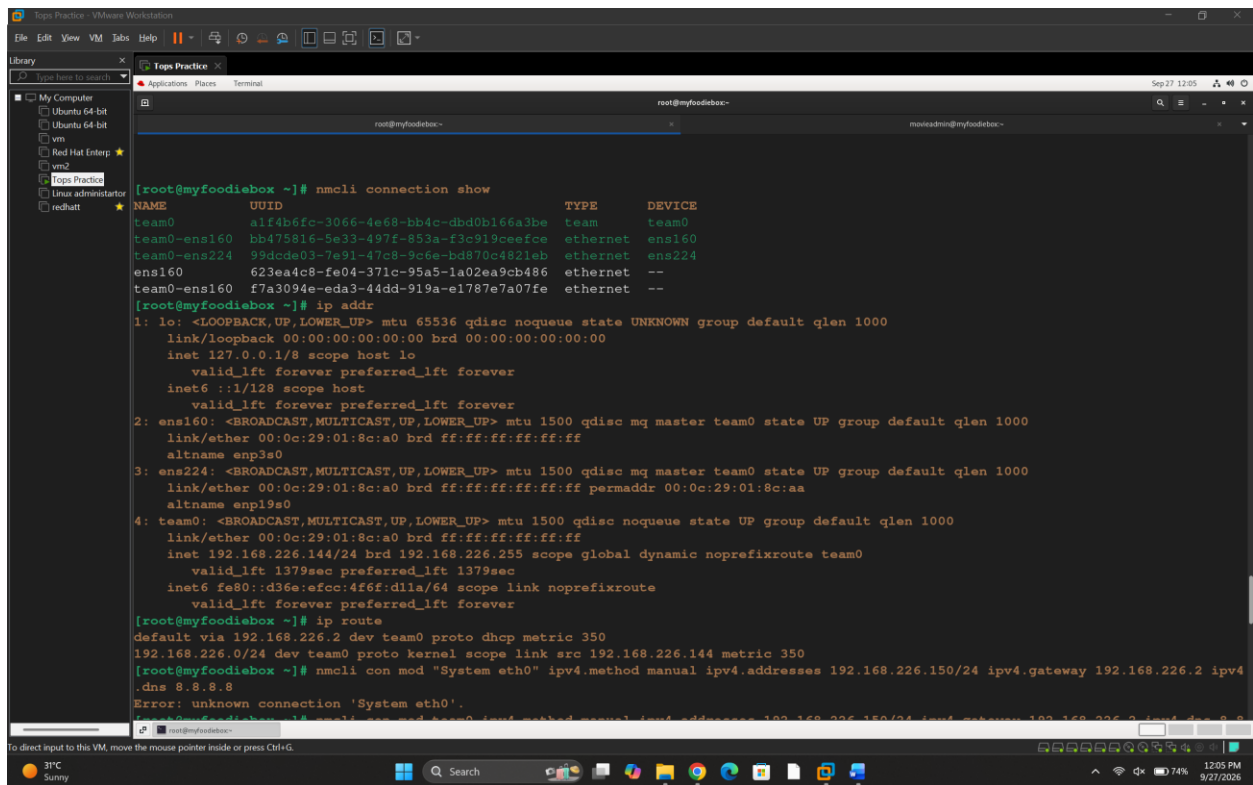
```
[root@myfoodiebox ~]# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq master team0 state UP group default qlen 1000
    link/ether 00:0c:29:01:8c:a0 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
3: ens224: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq master team0 state UP group default qlen 1000
    link/ether 00:0c:29:01:8c:a0 brd ff:ff:ff:ff:ff:ff permaddr 00:0c:29:01:8c:aa
    altname enp19s0
4: team0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 00:0c:29:01:8c:a0 brd ff:ff:ff:ff:ff:ff
    inet 192.168.226.144/24 brd 192.168.226.255 scope global dynamic noprefixroute team0
        valid_lft 1440sec preferred_lft 1440sec
    inet6 fe80::d36e:efcc:4f6f:d11a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@myfoodiebox ~]# ip -br addr
lo                UNKNOWN          127.0.0.1/8 ::1/128
ens160            UP
ens224            UP
team0             UP                192.168.226.144/24 fe80::d36e:efcc:4f6f:d11a/64
[root@myfoodiebox ~]#
```

2.

Configure a static IP address for your system using nmcli so that your device always gets the same IP (for example, 192.168.1.50/24 with gateway 192.168.1.1 and DNS 8.8.8.8).

Hint: Use 'nmcli con mod' to set ipv4.addresses, ipv4.gateway, and ipv4.dns, then bring the connection down and up to apply changes.

>

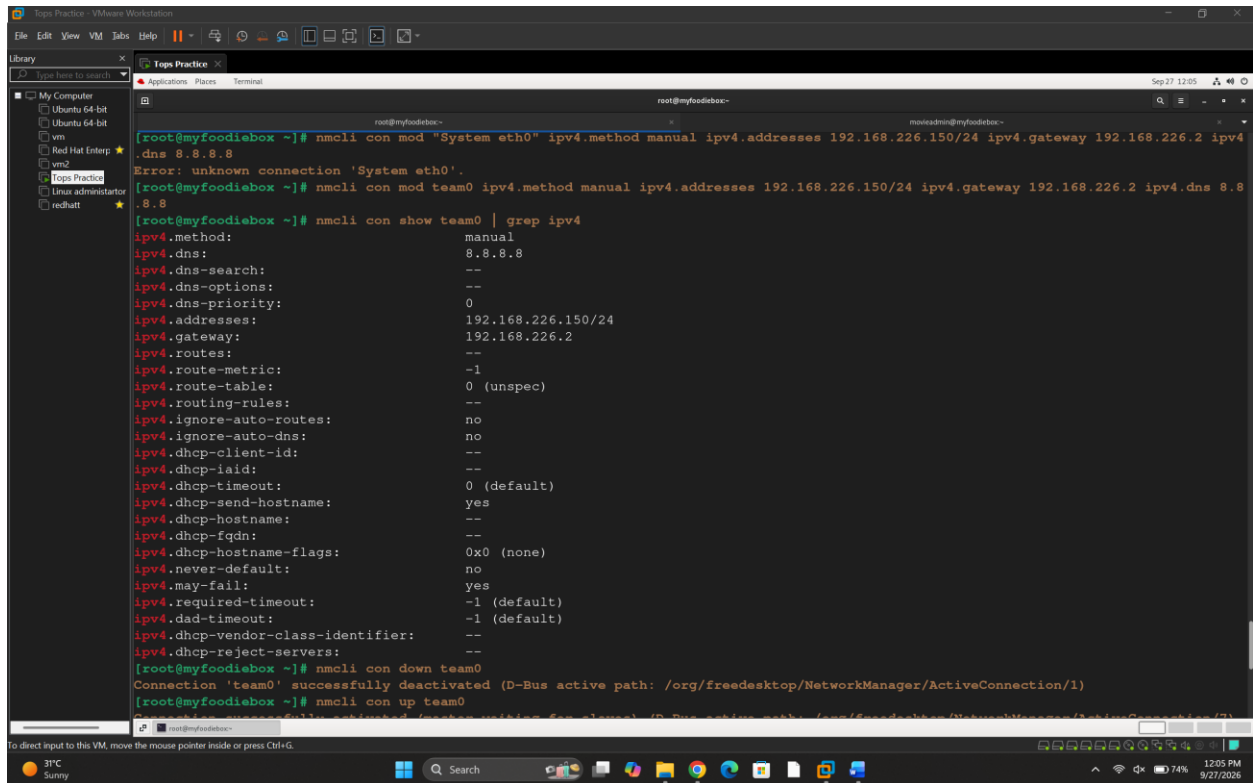


```
[root@myfoodiebox ~]# nmcli connection show
NAME                UUID                                  TYPE      DEVICE
team0               a1f4b6fc-3066-4e68-bb4c-dbd0b166a3be team       team0
team0-ens160        bb475816-5e33-497f-853a-f3c919ceefce ethernet  ens160
team0-ens224        99dcde03-7e91-47c8-9c6e-bd870c4821eb ethernet  ens224
ens160              623ea4c8-fe04-371c-95a5-1a02ea9cb486 ethernet  --
team0-ens160        f7a3094e-eda3-44dd-919a-e1787e7a07fe ethernet  --

[root@myfoodiebox ~]# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq master team0 state UP group default qlen 1000
    link/ether 00:0c:29:01:8c:a0 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
3: ens224: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq master team0 state UP group default qlen 1000
    link/ether 00:0c:29:01:8c:a0 brd ff:ff:ff:ff:ff:ff permaddr 00:0c:29:01:8c:aa
    altname enp19a0
4: team0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 00:0c:29:01:8c:a0 brd ff:ff:ff:ff:ff:ff
    inet 192.168.226.144/24 brd 192.168.226.255 scope global dynamic noprefixroute team0
        valid_lft 1379sec preferred_lft 1379sec
    inet6 fe80::d36e:efcc:4f6f:d11a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

[root@myfoodiebox ~]# ip route
default via 192.168.226.2 dev team0 proto dhcp metric 350
192.168.226.0/24 dev team0 proto kernel scope link src 192.168.226.144 metric 350

[root@myfoodiebox ~]# nmcli con mod "System eth0" ipv4.method manual ipv4.addresses 192.168.226.150/24 ipv4.gateway 192.168.226.2 ipv4.dns 8.8.8.8
Error: unknown connection 'System eth0'.
```



```
[root@myfoodiebox ~]# nmcli con mod "System eth0" ipv4.method manual ipv4.addresses 192.168.226.150/24 ipv4.gateway 192.168.226.2 ipv4.dns 8.8.8.8
Error: unknown connection 'System eth0'.

[root@myfoodiebox ~]# nmcli con mod team0 ipv4.method manual ipv4.addresses 192.168.226.150/24 ipv4.gateway 192.168.226.2 ipv4.dns 8.8.8.8

[root@myfoodiebox ~]# nmcli con show team0 | grep ipv4
ipv4.method:                manual
ipv4.dns:                    8.8.8.8
ipv4.dns-search:             --
ipv4.dns-options:            --
ipv4.dns-priority:           0
ipv4.addresses:              192.168.226.150/24
ipv4.gateway:                192.168.226.2
ipv4.routes:                 --
ipv4.route-metric:           -1
ipv4.route-table:            0 (unspec)
ipv4.routing-rules:          --
ipv4.ignore-auto-routes:     no
ipv4.ignore-auto-dns:        no
ipv4.dhcp-client-id:         --
ipv4.dhcp-iaid:              --
ipv4.dhcp-timeout:           0 (default)
ipv4.dhcp-send-hostname:     yes
ipv4.dhcp-hostname:          --
ipv4.dhcp-fqdn:              --
ipv4.dhcp-hostname-flags:    0x0 (none)
ipv4.never-default:          no
ipv4.may-fail:               yes
ipv4.required-timeout:       -1 (default)
ipv4.dad-timeout:            -1 (default)
ipv4.dhcp-vendor-class-identifier: --
ipv4.dhcp-reject-servers:    --

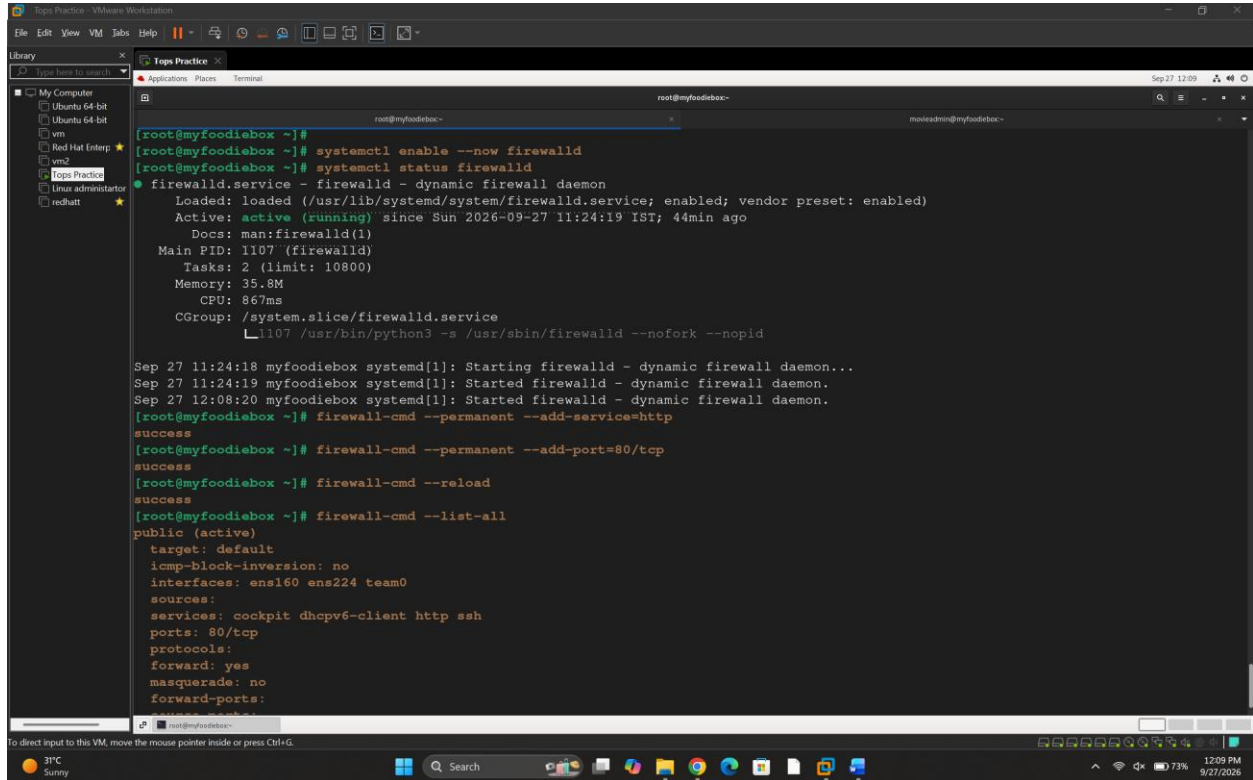
[root@myfoodiebox ~]# nmcli con down team0
Connection 'team0' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/1)

[root@myfoodiebox ~]# nmcli con up team0
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)
```

4.

Enable the firewall on your system using ufw, then allow incoming connections only on port 80 (HTTP) and block all others. Show the status output after applying these rules.

>



The screenshot shows a terminal window within a VMware Workstation environment. The terminal is running on a system named 'myfoodiebox'. The user is root. The following commands and outputs are shown:

```
[root@myfoodiebox ~]# systemctl enable --now firewalld
[root@myfoodiebox ~]# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2026-09-27 11:24:19 IST; 44min ago
     Docs: man:firewalld(1)
    Main PID: 1107 (firewalld)
      Tasks: 2 (limit: 10800)
     Memory: 35.8M
        CPU: 867ms
    CGroup: /system.slice/firewalld.service
            └─1107 /usr/bin/python3 -s /usr/sbin/firewalld --nofork --nopid

Sep 27 11:24:18 myfoodiebox systemd[1]: Starting firewalld - dynamic firewall daemon...
Sep 27 11:24:19 myfoodiebox systemd[1]: Started firewalld - dynamic firewall daemon.
Sep 27 12:08:20 myfoodiebox systemd[1]: Started firewalld - dynamic firewall daemon.
[root@myfoodiebox ~]# firewall-cmd --permanent --add-service=http
success
[root@myfoodiebox ~]# firewall-cmd --permanent --add-port=80/tcp
success
[root@myfoodiebox ~]# firewall-cmd --reload
success
[root@myfoodiebox ~]# firewall-cmd --list-all
public (active)
target: default
icmp-block-inversion: no
interfaces: ens160 ens224 team0
sources:
services: cockpit dhcpv6-client http ssh
ports: 80/tcp
protocols:
forward: yes
masquerade: no
forward-ports:
source-ports:
```

The bottom of the image shows the VMware Workstation interface with the 'myfoodiebox' VM selected in the library. The system tray at the bottom indicates a temperature of 31°C and a sunny weather condition.

```
Docs: man:firewalld(1)
Main PID: 1107 (firewalld)
Tasks: 2 (limit: 10800)
Memory: 35.8M
CPU: 867ms
CGroup: /system.slice/firewalld.service
└─1107 /usr/bin/python3 -s /usr/sbin/firewalld --nofork --nopid

Sep 27 11:24:18 myfoodiebox systemd[1]: Starting firewalld - dynamic firewall daemon...
Sep 27 11:24:19 myfoodiebox systemd[1]: Started firewalld - dynamic firewall daemon.
Sep 27 12:08:20 myfoodiebox systemd[1]: Started firewalld - dynamic firewall daemon.
[root@myfoodiebox ~]# firewall-cmd --permanent --add-service=http
success
[root@myfoodiebox ~]# firewall-cmd --permanent --add-port=80/tcp
success
[root@myfoodiebox ~]# firewall-cmd --reload
success
[root@myfoodiebox ~]# firewall-cmd --list-all
public (active)
target: default
icmp-block-inversion: no
interfaces: ens160 ens224 team0
sources:
services: cockpit dhcpv6-client http ssh
ports: 80/tcp
protocols:
forward: yes
masquerade: no
forward-ports:
source-ports:
icmp-blocks:
rich rules:
[root@myfoodiebox ~]#
[root@myfoodiebox ~]#
[root@myfoodiebox ~]#
```

5.

Use firewalld or iptables to temporarily block all outgoing traffic to a specific website (for example, www.spotify.com) and then verify that you cannot ping the site from your terminal. Hint: You may need to use the site's IP address instead of the domain name.

>

